Assignment - 3 - RTOS

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Task	Capacity	Period	Deadline
t_1	3	20	7
t_2	2	5	4
t_3	2	10	8

$(t_2 > t_3 > t_1)$

~~Q.1 Rate monotic scheduling (RMS~~

a) $LCM(20, 5, 10) = 20$

$U = 3/20 + 2/5 + 2/10 = 0.75$

Deadlines $= t_1 = 7$

$t_2 = 4, 9, 14, 17$

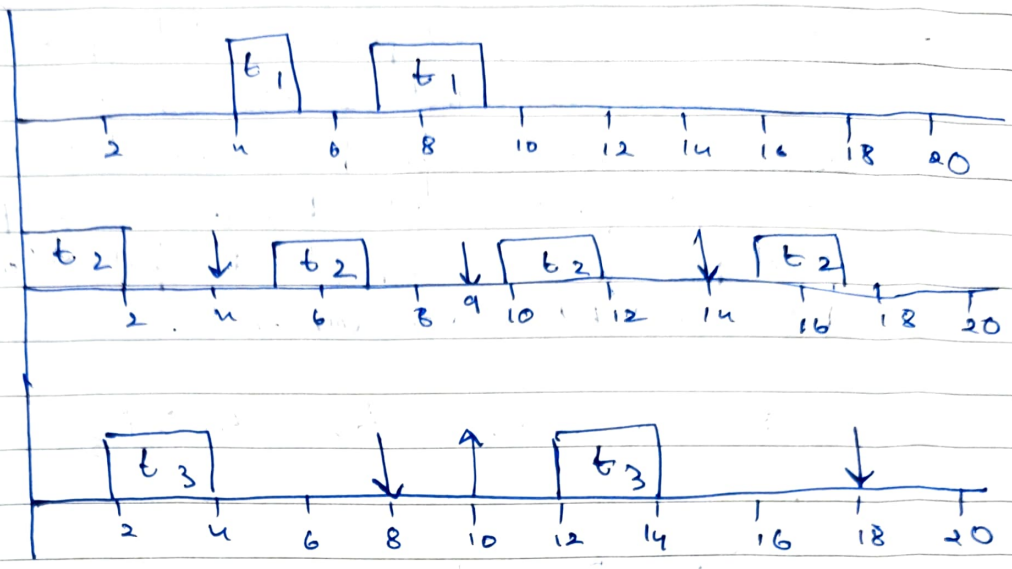
$t_3 = 8, 18$

1a) Rate monotonic Scheduling

RMS is a fixed priority preemptive algorithm the task with shortest period (highest rate of occurrence) gets the highest priority. Priorities are assigned once, offline and never change. Deadlines are not considered

$t_2 > t_3 > t_1 \rightarrow$ Priority order

Schedule



0 - 2 t_2 is highest

2 - 4 t_3 runs

4 - 5 → t_1 gets it's turn

5 - 7 → t_2 2nd job preempts it

7 - 9 t_1 resumes and finishes at 7

10 - 12 t_2

12 - 14 t_3

15 - 17 t_2

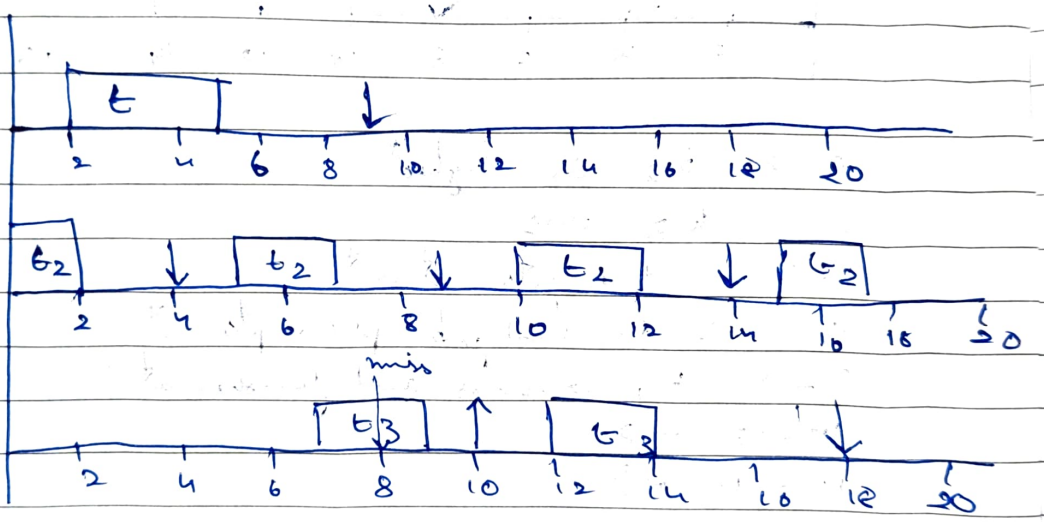
Idle is 9 - 10
14 - 15 → 17 - 20
} Idle

Why it fails is before time 7, t_1 only get 1 time unit (4 - 5). So, it's 2 units short t_2 and 3 keeps preempting it because RMS by period. And its period (10) is longer than D < T, RMS can be wrong choice

20) Deadline monotonic scheduling (DMS)

→ DMS is also fixed priority, but priority goes to the task with closest relative deadline. It is better than RMS when $D < T$. In that case RMS is not optimal, while DMS is the optimal fixed priority scheme. When D for all tasks DMS and RMS are identical.

Priority order $t_2 > t_1 > t_3$



2-5 t_1 runs fully and finished at 5 deadline at 7

5 → t_2 2nd job

7-9 t_3 runs and finish at 9

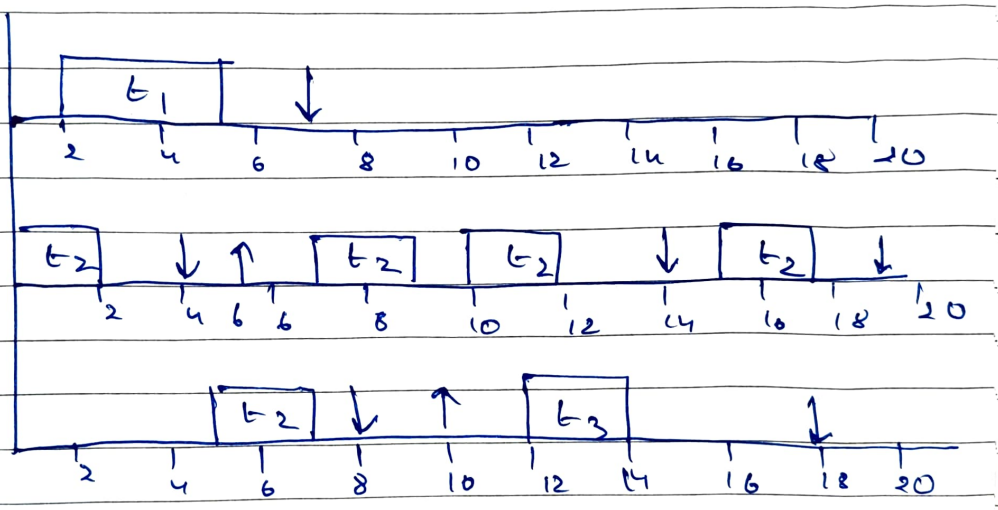
10-12 t_2

12-14 t_3 2nd job deadline 14

t_3 has the lowest priority t_2 and t_1 will run

Q3) Earliest deadline first

→ EDF is dynamic priority algorithm. It selects during each scheduling point (job release or completion) the ready job with the nearest absolute deadline. It runs in RM and DMS. A task's priority is fixed. In EDF the same task can be highest priority at one and lowest at other because priority of the job absolute deadline.



At t=0 t2 - t2 is scheduled as it gets the highest priority because of its deadline being the closest

At t2 t=4 t=5 t1 is picked and scheduled as it gets the highest priority since its deadline is closer among t1 and t3

At t=5 t=7 = t2 is scheduled even though t2 is ready because its deadline is closest among t2 and t2 (8 & 9). So t3 gets the highest priority

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D) In EDF every task meets its deadline. This is because priorities are dynamically assigned based on the closest deadline.